CHERN CLASSES ARE CENTRAL AND COMMUTE WITH BIVARIANT CLASSES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's parenthetical at source line 7624, '(calculation omitted)', is retained verbatim inside the counted proof; the reduction and the identification by uniqueness are inside the span.

Allowed standard prerequisites (editorial).

- The characterisation of the Chern classes by one relation on the projective bundle. Same chapter chow.tex, label lemma-determine-intersections, invoked at proof lines 7604 and 7626. Gives the relation satisfied by the classes obtained by capping with the Chern classes of the bundle, and then identifies the classes obtained after applying the bivariant class with the caps of the Chern classes of the pulled back bundle.
- Centrality of the first Chern class. Same chapter chow.tex, label lemma-c1-center, invoked at proof line 7613. Lets the bivariant class be moved across the powers of the first Chern class of the twisting sheaf in the displayed computation.
- The properties of a bivariant class. Same chapter chow.tex, definition of a bivariant class used at proof lines 7613-7622. Allows the bivariant class to be applied to the relation, to commute with the flat pullback along the projection from the base changed projective bundle, and to be moved past the cap.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0B7B: sources/stacks-chow/pinned-original-source.tex, source lines 6332–6332. Navigation pointer for \ref{lemma-c1-center} used inside the counted span (line 7613); the argument that consumes it is retained in the span.

Source reference. Original lemma 02U6: sources/stacks-chow/pinned-original-source.tex, source lines 7280–7280. Navigation pointer for \ref{lemma-determine-intersections} used inside the counted span (line 7604); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 7582); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of rank r. Then*

1. *$c_j(\mathcal{E}) \in A^j(X)$ is in the center of $A^*(X)$ and*
2. *if $f : X' \to X$ is locally of finite type and $c \in A^*(X' \to X)$, then $c \circ c_j(\mathcal{E}) = c_j(f^*\mathcal{E}) \circ c$.*

In particular, if $\mathcal{F}$ is a second locally free $\mathcal{O}_X$-module on X of rank s, then

$$c_i(\mathcal{E}) \cap c_j(\mathcal{F}) \cap \alpha = c_j(\mathcal{F}) \cap c_i(\mathcal{E}) \cap \alpha$$

as elements of $\mathrm{CH}_{k-i-j}(X)$ for all $\alpha \in \mathrm{CH}_k(X)$.

Proof. It is immediate that (2) implies (1). Let $\alpha \in \mathrm{CH}_k(X)$. Write $\alpha_j = c_j(\mathcal{E}) \cap \alpha$, so $\alpha_0 = \alpha$. By Lemma 02U6 we have

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi^*(\alpha_{r-i}) = 0$$

in the chow group of the projective bundle $(\pi : P \to Y, \mathcal{O}_P(1))$ associated to $\mathcal{E}$. Denote $\pi' : P' \to X'$ the base change of π by f. Using Lemma 0B7B and the properties of bivariant classes we obtain

$$\begin{aligned} 0 &= c \cap \Big(\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi^*(\alpha_{r-i}) \Big) \\ &= \sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_{P'}(1))^i \cap (\pi')^*(c \cap \alpha_{r-i}) \end{aligned}$$

in the Chow group of P' (calculation omitted). Hence we see that $c \cap \alpha_j$ is equal to $c_j(f^*\mathcal{E}) \cap (c \cap \alpha)$ by the characterization of Lemma 02U6. This proves the lemma. □